Yongbo Chen

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Computer Science Ph.D. student working at the intersection of human-computer interaction, visualization, and software engineering. My research combines qualitative studies, interactive prototyping, and program analysis to help people understand and improve software and data workflows.

Education

Tulane University Aug. 2024 – Present
Ph.D. in Computer Science, expected 05/2029 | GPA: 3.975/4.00
Advisor: Rebecca Faust

University of California, Irvine Sep. 2022 – Jun. 2024
M.S. in Software Engineering | GPA: 3.94/4.00

The Ohio State University Aug. 2018 – May 2022
B.S. in Computer Science and Engineering

Selected Publications

- **Y. Chen**, Y. Zhu, and R. Faust. *Debugging as Evidence-Driven Reasoning: Visualization Opportunities in Data-Intensive Programming*. **IEEE VIS 2026**, short paper, to appear. [Paper]
- Y. Zhu, **Y. Chen**, and R. Faust. *KeySI: An Interaction Framework for Tuning Text Embeddings Based on Human Feedback*. **IEEE VIS 2026**, full paper, to appear. [Paper]
- Y. He, **Y. Chen**, J. Ayala, Y. Zhou, Q. Wang, and J. Garcia. *Breaking Strong Encapsulation: A Comprehensive Study of Java Module Abuse*. **ICSE 2026**, Research Track. [Publication]

Research Experience

Ph.D. Research | Tulane University Aug. 2024 – Present
Advisor: Rebecca Faust

Visual Support for Debugging Data Workflows

- Conducted a qualitative interview study with 9 participants to understand debugging in data-intensive programming, including pandas and computational notebooks. Characterized how practitioners identify discrepancies between expected and observed program behavior.
- Derived visualization design requirements for aligning evidence across scattered artifacts, comparing expected and observed states, and tracing changes across pipeline stages; first-author short paper accepted at IEEE VIS 2026.
- Designed a two-session study combining interviews and design probes to investigate how users express expectations about program behavior; currently prototyping a visual interface for side-by-side comparison of expected and observed states.

KeySI: Human Feedback for Text Embedding Refinement

- Co-authored an interactive framework for tuning text embeddings through human feedback (second author, IEEE VIS 2026 full paper).
- Contributed to study design, conducted pilot sessions, and supported the user study and interface evaluation.

Research Experience (continued)

Graduate Student Researcher | UC Irvine Apr. 2023 – May 2024

Advisor: Joshua Garcia

Java Module Encapsulation: Empirical Research and Tool Development

- Studied how Java projects bypass module boundaries to access internal APIs. Built a GitHub crawler collecting 4,079 candidate issues and used qualitative coding, with each issue independently labeled by two authors, to characterize module abuse; co-authored the resulting ICSE 2026 paper.
- Designed and built **BEAD**, a static-analysis tool that detects access to JDK-internal APIs through direct calls and reflection, using JavaParser, Soot, call graphs, and dataflow analysis.
- Detected and manually confirmed **187 encapsulation violations across 5 real-world applications** under JDK 17 (182 direct calls and 5 reflective calls).
- Investigated module decay and API over-exposure in evolving Java frameworks using the DARCY approach, including comparative analysis of framework architectures and module calls.

Master's thesis: *Automatic Detection of Breaking Strong Encapsulation in Java Modules* (2024).

Selected Projects

CodeMentor: AI-Assisted Programming Education Sep. 2024 – Dec. 2024

- Conducted interviews and surveys, analyzed learning needs through thematic analysis, and developed personas, task scenarios, and prototypes for an LLM-integrated programming tutor.
- Led heuristic evaluation and usability testing to refine the design; implemented a Node.js prototype with lesson navigation, adaptive chat tutoring, and progress tracking.

Accessibility Impacts of Dark Patterns Jan. 2023 – Mar. 2023

- Examined 135 Android apps and recorded 459 dark patterns across five categories; analyzed Google TalkBack behavior to classify potential impacts on blind users into three levels.

Project Lily: Civic Engagement Platform Jan. 2022 – May 2022

Undergraduate capstone | Backend engineer on a three-person backend team

- Built Spring Boot microservices for a platform modernization, owning email-status tracking with AWS SNS/SQS and automated reminders; integrated SMS and email delivery with a PostgreSQL database.
- Implemented asynchronous processing and exception handling; verified backend functionality with JUnit and Postman, with test coverage measured using JaCoCo.

Skills

Research methods: Semi-structured interviews, surveys, thematic analysis, study design, design probes, usability testing, heuristic evaluation.

Programming: Python, Java, JavaScript, SQL, C, C#, MATLAB, Ruby, HTML.

Visualization & prototyping: D3.js, pandas, Jupyter notebooks, Node.js, Unity 3D.

Program analysis & engineering: Soot, JavaParser, static and dataflow analysis, Git, Spring Boot, AWS (SNS/SQS, RDS), PostgreSQL.

Teaching & Service

Teaching Assistant, Introduction to Computer Science II (CMPS 1600), Tulane University. Fall 2026

Junior Reviewer, IEEE VIS 2026.